

Kaiqi Yang

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Research Interests

- Applications of Large Language Models; Conversational AI; Computational Social Science

Education

- **Michigan State University** 2022 – 2027 (expected)
PhD Candidate, Computer Science and Engineering
◦ Advisor: Dr. Hui Liu, Dr. Jiliang Tang East Lansing, USA
- **Fudan University** 2019 – 2021
Master of Applied Statistics, School of Data Science Shanghai, China
- **Fudan University** 2015 – 2019
Bachelor of Sociology, Department of Sociology Shanghai, China

Experiences

- **ByteDance Inc.** May 2026 – Aug 2026
Research Scientist Intern, Ark_AML San Jose, CA
 - Developed an LLM-agent harness framework for evolutionary skill optimization, supporting relevant-skill retrieval, skill performance attribution, and performance signal fusion.
 - Designed a graph-based dependency modeling method to capture interactions among evaluation dimensions, improved the consistency and interpretability of automated evaluation.
 - Built automated evaluation generation pipeline for optimizing agent skills, enlarge the supervision signals to help RL.
- **Amazon Inc.** Aug 2025 – Nov 2025
Applied Scientist Intern Seattle, USA
 - Research Topics: Foundational AI for Stores; Conversational Generation and Evaluation
- **TAL Education Group** Jun 2024 – Aug 2024
Summer Intern Beijing, China
 - Research Topics: Applications of Large Language Models; Intelligent Tutoring Systems for Elementary Math
- **Data Science and Engineering Lab, Michigan State University** Aug 2022 – Present
Research Assistant, Department of Computer Science and Engineering East Lansing, USA
 - Research Areas: Large Language Models, AI for Education, Computational Social Science
 - Advisors: Dr. Jiliang Tang and Dr. Hui Liu
- **Department of Computing, The Hong Kong Polytechnic University** May 2023 – Aug 2023
Research Intern Hong Kong
 - Research Topics: Large Language Models; Graph Data Mining

Publications

* indicates equal contribution.

- **Think-Before-Speak: From Internal Evaluation to Public Expression in Multi-Agent Social Simulation**
Kaiqi Yang, Tai-Quan Peng, Sanguk Lee, Hui Liu
Presented at KDD'26 Workshop on "Agentic AI for Scientific and Societal Advances", 2026.
- **Exploring Prompt Injection Attacks on LLM-Based Automatic Grading Systems**
Hang Li*, Fedor Filippov*, Yuling Lin, Pengfei He, **Kaiqi Yang**, Yucheng Chu, Yingqian Cui, Hui Liu, Jiliang Tang
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.
- **Exploring Solution Divergence and Its Effect on Large Language Model Problem Solving**
Hang Li, **Kaiqi Yang**, Yucheng Chu, Hui Liu, Jiliang Tang
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2026.
- **Confusion-Aware Rubric Optimization for LLM-based Automated Grading**
Yucheng Chu, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Joseph Krajcik, Namsoo Shin, Jiliang Tang
International Conference on Educational Data Mining (EDM), 2026.
- **From Flat to Structural: Enhancing Automated Short Answer Grading with GraphRAG**
Yucheng Chu, Haoyu Han, Shen Dong, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Joseph Krajcik, Namsoo Shin, Hui Liu
International Conference on Educational Data Mining (EDM), 2026.
- **A LLM-Driven Multi-Agent Systems for Professional Development of Mathematics Teachers**
Kaiqi Yang*, Hang Li*, Yucheng Chu, Ahreum Han, Yasemin Copur-Gencturk, Hui Liu
International Conference on Artificial Intelligence in Education (AIED), 2026.
- **Optimizing In-Context Demonstrations for LLM-based Automated Grading**
Yucheng Chu, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Kevin Haudek, Joseph Krajcik, Jiliang Tang
International Conference on Artificial Intelligence in Education (AIED), 2026.
- **Designing Professional Development in Mathematics Teacher Education with AI as a Facilitator: A Roadmap Grounded in High-Quality Professional Development Features**
Nickolina Yankova, Jinhyo Cho, Hang Li, **Kaiqi Yang**, Yasemin Copur-Gencturk, Jiliang Tang
Journal of Mathematics Teacher Education, 2026.
- **Beyond Partisan Leaning: A Comparative Analysis of Political Bias in Large Language Models**
Tai-Quan Peng, **Kaiqi Yang**, Sanguk Lee, Hang Li, Yucheng Chu, Yuping Lin, Hui Liu
Journal of Information Technology & Politics, 2026.
- **LLM-based Automated Grading with Human-in-the-Loop**
Hang Li*, Yucheng Chu*, **Kaiqi Yang**, Yasemin Copur-Gencturk, Jiliang Tang
IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE), 2025.
- **Iterative LLM-Based Generation and Refinement of Distracting Conditions in Math Word Problems**
Kaiqi Yang, Hang Li, Yucheng Chu, Zitao Liu, Mi Tian, Hui Liu
IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE), 2025.

- [Ask-Before-Detection: Identifying and Mitigating Conformity Bias in LLM-Powered Error Detector for Math Word Problem Solutions](#)
Hang Li, Tianlong Xu, **Kaiqi Yang**, Yucheng Chu, Yanling Chen, Yichi Song, Qingsong Wen, Hui Liu
Annual Meeting of the Association for Computational Linguistics (ACL), 2025.
- [A LLM-Powered Automatic Grading Framework with Human-Level Guidelines Optimization](#)
Yucheng Chu, Hang Li, **Kaiqi Yang**, Harry Shomer, Hui Liu, Yasemin Copur-Gencturk, Jiliang Tang
International Conference on Educational Data Mining (EDM), 2025.
- [Are Large Language Models \(LLMs\) Good Social Predictors?](#)
Kaiqi Yang, Hang Li, Hongzhi Wen, Tai-Quan Peng, Jiliang Tang, Hui Liu
Conference on Empirical Methods in Natural Language Processing (EMNLP), 2024.
- [Content Knowledge Identification with Multi-Agent Large Language Models \(LLMs\)](#)
Kaiqi Yang, Yucheng Chu, Taylor Darwin, Ahreum Han, Hang Li, Hongzhi Wen, Yasemin Copur-Gencturk, Jiliang Tang, Hui Liu
International Conference on Artificial Intelligence in Education (AIED), 2024.
- [Spectral-Aware Augmentation for Enhanced Graph Representation Learning](#)
Kaiqi Yang, Haoyu Han, Wei Jin, Hui Liu
ACM International Conference on Information and Knowledge Management (CIKM), 2024.

Preprints

* indicates equal contribution.

- [How Uncertain Is the Grade? A Benchmark of Uncertainty Metrics for LLM-Based Automatic Assessment](#)
Kaiqi Yang*, Hang Li*, Xianxuan Long, Fedor Filippov, Yucheng Chu, Yasemin Copur-Gencturk, Peng He, Cory Miller, Namsoo Shin, Joseph Krajcik, Hui Liu, Jiliang Tang
arXiv preprint arXiv:2602.16039, 2026.
- [Enhancing LLM-Based Short Answer Grading with Retrieval-Augmented Generation](#)
Yucheng Chu, Peng He, Hang Li, Haoyu Han, **Kaiqi Yang**, Yu Xue, Tingting Li, Joseph Krajcik, Jiliang Tang
arXiv preprint arXiv:2504.05276, 2025.
- [Exploring Social Desirability Response Bias in Large Language Models: Evidence from GPT-4 Simulations](#)
Sanguk Lee*, **Kaiqi Yang***, Tai-Quan Peng, Ruth Heo, Hui Liu
arXiv preprint arXiv:2410.15442, 2024.

Presentations

- [Bringing Generative Artificial Intelligence \(GenAI\) to Education](#)
Hang Li, **Kaiqi Yang**, Yucheng Chu, Jiliang Tang
Tutorial, International Conference on Educational Data Mining (EDM), Palermo, Italy, 2025.
- [Innovative Artificial Intelligence Solutions for Improving Online Professional Development for Teachers: Multi-Agent Large Language Models](#)
Kaiqi Yang, Yasemin Copur-Gencturk, Ahreum Han, Hang Li, Jingxian Li, Jiliang Tang
Symposium: "The Use of AI in Education and Research in Meaningful and Responsible Ways," Annual Meeting of the American Educational Research Association (AERA), Denver, USA, 2025.

Honors and Awards

- Student Travel Award, ACM CIKM 2024
- Outstanding Graduate Student Scholarship, School of Data Science, Fudan University 2020/21
- Outstanding Undergraduate Student Scholarship, Fudan University 2016/17/18/19

Teaching

- Teaching Assistant, CSE 482: Big Data Analysis Spring 2025, Spring 2024
Responsibilities included grading coursework, holding office hours, and mentoring students on honors projects.
- Teaching Assistant, CSE 440: Introduction to Artificial Intelligence Fall 2024, Fall 2023
Responsibilities included managing online discussions, holding office hours, and grading coursework.

Mentoring

- Yucheng Chu, PhD Student Dec 2023 – Jan 2025
Project: Large Language Models for STEM Education
Co-authored publication: AIED 2024

Service

- Conference Reviewer 2022 – Present
Reviewed for WWW, WSDM, NeurIPS, ICLR, ICDM, ICML, ACL
- Journal Reviewer 2022 – Present
Reviewed for TKDE, TNNLS, TKDD